CJC Open 7 (2025) 795–798

Original Article

Echoing Intelligence: Assessing the Capabilities of a Novel Generative AI With a Large Language Model in Cardiology Education

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ABSTRACT

Background: Artificial intelligence (AI) holds promise for enhancing medical education, particularly in complex fields like cardiology. We assessed the ability of a large language model (LLM) to generate and evaluate educational material comparable to that created by human experts.

Methods: We trained an AI model on cardiology-specific content using 80 lectures from the St. Michael's Hospital Virtual Echo Rounds. The AI generated 10 multiple-choice questions (MCQs), and experienced cardiologists crafted an additional 10 MCQs. Eleven postgraduate year 4-6 cardiology trainees answered all 20 questions and attempted to identify the source (AI or human) of each question. The AI also answered the same set of questions. We analyzed performance using the Wilcoxon signed-rank test and recognition ability.

Results: Trainees scored similarly on AI-generated and human-generated questions (median 8/10 vs 8/10; $P > 0.05$). Their ability to identify the source of questions did not exceed chance levels (median correct identifications: 10/20; $P > 0.05$). The AI achieved 95% accuracy on AI-generated questions and 100% on human-generated questions.

Conclusions: The AI-generated educational content was of comparable quality to that produced by human experts, and trainees could not reliably distinguish between the 2 sources. Our findings suggest that AI could significantly augment cardiology education by providing high-quality, scalable learning resources.

RÉSUMÉ

Contexte : L'intelligence artificielle (IA) est prometteuse lorsqu'il s'agit d'améliorer la formation médicale, en particulier dans les domaines complexes comme la cardiologie. Nous avons évalué la capacité d'un grand modèle de langage (LLM pour *Large Language Model*) à générer et à évaluer du matériel pédagogique comparable à celui créé par des experts humains.

Méthodologie : Nous avons entraîné un modèle d'IA sur du contenu spécifique de la cardiologie en utilisant 80 conférences tirées des rondes virtuelles en échocardiographies (*Virtual Echo Rounds*) du St. Michael's Hospital. L'IA a généré 10 questions à choix multiples, tandis que des cardiologues expérimentés en ont élaboré 10 autres. Onze stagiaires en cardiologie (années d'étude 4 à 6) ont répondu aux 20 questions en tentant d'identifier la source (IA ou humaine) de chaque question. L'IA a également répondu au même jeu de questions. Nous avons analysé les performances à l'aide du test de Wilcoxon à rangs signés et de la capacité de reconnaissance.

Résultats : Les stagiaires ont obtenu des scores similaires aux questions générées par l'IA et à celles formulées par l'être humain (médiane de 8/10 *vs* 8/10; $p > 0{,}05$). Leur capacité à identifier la source des questions n'a pas dépassé le hasard (identifications correctes médianes : 10/20; $p > 0{,}05$). L'IA a obtenu un score de 95 % pour les questions générées par l'IA et un score de 100 % pour celles générées par l'être humain.

Conclusions : Le contenu éducatif généré par l'IA était d'une qualité comparable à celui produit par des experts humains, et les stagiaires ne sont pas parvenus à distinguer les deux sources de manière fiable. Nos résultats indiquent que l'IA pourrait grandement contribuer à la formation en cardiologie en fournissant des ressources d'apprentissage modulables de haute qualité.

Advancements in artificial intelligence (AI) have opened new avenues for enhancing medical education. In cardiology—a field characterized by rapidly expanding knowledge and complex clinical scenarios—there is a pressing need for innovative educational tools that can adapt to individual learning needs. Traditional educational methods may struggle to keep pace with the evolving

Received for publication January 4, 2025. Accepted March 16, 2025.

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See page 798 for disclosure information.

https://doi.org/10.1016/j.cjco.2025.03.013

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landscape, highlighting the potential role of AI in addressing this gap.

Previous studies have demonstrated AI's capacity to personalize educational material and improve learner engagement. Wartman and Combs[1] emphasized the transition from information dissemination to intelligent, adaptive learning systems in medical education. Masters underscored the practical framework AI provides, facilitating critical thinking and application.[2] Moreover, Narayanan et al.[3] highlighted AI's ability to address individual knowledge gaps and offer real-time feedback, enhancing learning outcomes. Cheung et al.[4] found that AI-generated examination questions matched the quality of those created by human experts, even in high-stakes settings like medical licensing examinations. Additionally, Chan and Zary[5] discussed how AI tutors could revolutionize medical education by providing personalized learning experiences. Liu et al.[6] explored the role of AI in medical education, highlighting its potential to transform traditional teaching methods.

Canadian-specific educational frameworks and guidelines have also evolved to reflect the need for innovation. The Royal College of Physicians and Surgeons of Canada emphasizes competency-based medical education, and the Canadian Cardiovascular Society (CCS) provides nationally recognized guidelines and standards for training and practice.[7,8] AI-based educational tools may align with these principles by supporting learners as they progress toward milestones and competencies laid out by these Canadian bodies. Such tools could strengthen adherence to Canadian guidelines in areas like heart failure management, valvular disease, and imaging protocols and ensure consistent, high-quality instruction across diverse training sites in Canada.

Building on this foundation, our study aimed to evaluate the effectiveness of a large language model (LLM) in generating and assessing cardiology educational content. By comparing trainee performance on AI-generated vs human-generated questions and assessing their ability to distinguish between them, we sought to understand AI's potential role in advancing cardiology education.

Methods

Study design

We conducted an exploratory study involving 11 cardiology trainees from a Canadian university—affiliated training program. Participation was voluntary, and all participants provided informed consent privately. We conducted the study in accordance with the principles of the Declaration of Helsinki. Formal institutional ethics approval was not obtained, as the study was considered an educational quality improvement initiative that did not involve sensitive personal data or interventions affecting patient care.

Data source and AI training

We used the Podcast AI platform to transcribe 80 publicly available echo rounds (St. Michael's Hospital Virtual Echo Rounds) for their comprehensive coverage of echocardiographic and cardiovascular topics.[7] A research assistant manually reviewed these transcripts for major errors to ensure accuracy of medical terminology. We trained a GPT-4—based LLM on these verified transcripts. The LLM restricted itself from having real-time access to the Internet or external databases during question generation; it operated on a closed data set. We used reinforcement learning techniques to optimize question quality based on relevance, difficulty, and alignment with educational objectives.

Participants

Eleven participants in this study were postgraduate year 4-6 cardiology trainees in a Canadian university—affiliated program. They had each regularly attended echocardiography teaching sessions. Inclusion criteria included cardiology training program enrollment and informed consent provision. We aimed for a pilot, all-comers approach. Because the total available trainees rotating through echocardiography during the time of our study was 11, we included all volunteers for a sample of convenience. This approach, typical of preliminary feasibility studies, allowed us to gauge the viability of AI-generated questions and compare them to expert-generated questions without a formal power calculation.

Intervention

We administered 20 multiple-choice questions—10 generated by our LLM AI and 10 crafted by experienced cardiologists. Trainees answered all 20 questions and attempted to identify whether each question was generated by the AI or by a human. The specific AI-generated and human-generated questions are available in Supplemental Table S1.

Outcomes

The primary outcomes were as follows:

- Trainee performance on AI-generated vs human-generated questions
- Trainees' ability to correctly identify the source of each question

Secondary outcomes included the AI's performance on both sets of questions.

Statistical analysis

We compared individual trainee scores for AI-generated and human-generated questions using the Wilcoxon signed-rank test, which is appropriate for paired, nonparametric data. We chose this test because of the small sample size and the non-normal distribution of score differences. The test assumes that differences between paired observations are symmetrically distributed around the median.

For the Wilcoxon signed-rank test comparing trainee scores, n represents the number of nonzero differences (in this case, $n = 8$ trainees with nonzero differences). We calculated the exact P value to determine statistical significance.

We assessed trainees' ability to recognize the source of questions using the same test, comparing their recognition rates against a chance level of 50%, here, $n = 8$ trainees with nonzero differences from chance recognition rates. We set statistical significance at $P < 0.05$. We performed analyses using R, version 4.0.3 (R Foundation for Statistical Computing, Vienna, Austria).

Results

Trainee performance

Trainees achieved a median (interquartile range [IQR]) score of 8 (7-9) out of 10 on the AI-generated questions and 8 (7-8) out of 10 on the human-generated questions (Tables 1 and 2). The Wilcoxon signed-rank test showed no significant difference between the 2 sets of scores ($T = 13$; n = 8; $P = 0.65$), indicating comparable performance. Trainees correctly identified the question source (AI vs human) only 51.4% of the time (median [IQR], 10 [8-12] of 20), which did not exceed chance levels ($T = 18$; n = 9; $P = 0.68$; Table 3). Our analysis revealed no significant bias toward overcalling questions as either AI-generated or human-generated (50% of AI questions overcalled as human and 47% of human questions overcalled as AI).

AI performance

The AI achieved the following:

- Accuracy on AI-generated questions: 95% (9 of 10 correct)
- Accuracy on human-generated questions: 100% (10 of 10 correct)
- Correct identification of question source: 55% (11 of 20)

Discussion

Our findings demonstrate that an AI model can generate high-quality educational content for cardiology trainees, yielding scores comparable to those obtained from human-generated questions. Although we included only 11 participants, this pilot sample provided preliminary evidence that AI-generated questions can effectively assess knowledge and clinical reasoning in complex medical domains. These results align with previous work such as Cheung et al.,[4] who reported that AI-generated examination questions matched the quality of those created by human experts.

The inability of trainees to distinguish between AI-generated and human-generated questions highlights the sophistication of the AI's language and content generation capabilities. This finding suggests that AI-generated content can be seamlessly integrated into existing educational frameworks without compromising the authenticity of learning materials. Similarly, Hashimoto et al. 8 discussed how AI systems could generate clinically relevant content that meets educational standards.

Table 1. Individual trainee scores

Trainee	AI score	Human score	Difference (AI − human)
Trainee 1	9	9	0
Trainee 2	9	8	+1
Trainee 3	9	8	+1
Trainee 4	7	8	−1
Trainee 5	6	8	−2
Trainee 6	6	7	−1
Trainee 7	6	5	+1
Trainee 8	8	6	+2
Trainee 9	8	8	0
Trainee 10	7	7	0
Trainee 11	8	6	+2

AI, artificial intelligence.

Table 2. Total examination scores

Raw scores	AI-generated questions	Human-generated questions	Total score
Cardiology trainee performance (11 trainees)	83/110 (75)	80/110 (73)	163/220 (75)
PODCASTAI performance	9/10 (90)	10/10 (100)	19/20 (95)

Values within parentheses are percentages.
AI, artificial intelligence.

Our results have particular relevance in the Canadian context. As cardiology training programs nationwide aim to maintain high standards of teaching, AI-driven educational tools could provide a uniform baseline of complexity and content quality, regardless of geographic location or institutional resources. By aligning AI-generated content with Canadian Cardiovascular Society guidelines and the Royal College's competency-based frameworks, educators can ensure trainees receive consistent instruction that reflects national priorities in cardiovascular health care.

The AI's high performance in answering both AI-generated and human-generated questions underscores its robust comprehension of cardiology concepts. This dual capability suggests potential applications beyond content generation, such as serving as an intelligent tutor providing immediate feedback and personalized learning pathways. By leveraging AI's adaptability, educators can tailor educational experiences to individual trainee needs, potentially enhancing learning efficiency and outcomes.[5]

Implications for medical education

Integrating AI into medical education could address several challenges. AI enhances scalability by efficiently generating extensive educational content, alleviating some of the burdens on educators. Through adaptive learning algorithms, AI achieves personalization by tailoring material to individual trainee strengths and weaknesses.[6] AI improves consistency in educational content, ensuring uniform quality and difficulty across different cohorts.

In the Canadian setting, this consistency extends to aligning educational content with national standards. The Canadian health care environment, with its diverse patient populations and wide geographic distribution of trainees, could benefit from AI tools that deliver standardized, guideline-based educational content. This approach ensures that all Canadian trainees, whether in large urban centers or

Table 3. Recognition scores

Raw scores	AI-generated questions	Human-generated questions	Total correct recognition of AI vs human
Cardiology trainee performance (11 trainees)	55/110 (50)	58/110 (53)	113/220 (51)
PODCASTAI performance	1/10 (10)	10/10 (100)	11/20 (55)

Values within parentheses are percentages.
AI, artificial intelligence.

more remote regions, have equitable access to high-quality cardiology education.

By incorporating AI-generated content, educators may focus more on mentoring and direct clinical teaching, areas where human expertise is irreplaceable. This balance could enrich the educational environment, combining the efficiency of AI with the nuanced guidance of experienced clinicians.

Limitations

Our study has limitations that warrant consideration. We conducted this as a feasibility study with a relatively small sample (n = 11), which may limit generalizability. Larger, multicenter studies are needed to confirm these results and explore potential variability across different training programs. Additionally, we restricted our scope to multiple-choice questions derived from echocardiography lectures; expanding question formats to include open-ended cases or clinical simulations could offer further insight into the AI's versatility. Lastly, although we followed up on short-term learning outcomes, we did not assess long-term knowledge retention or the application of AI-generated content in real-world clinical decision making—key areas for future research.

Future directions

Future research should explore the integration of AI-generated content into the broader Canadian medical curriculum, including other training programs and specialties. Assessing the impact of these tools on training standards set by Canadian governing bodies—such as the Royal College—and evaluating how well they align with CCS guidelines could further inform the development of AI-driven educational platforms across the country.

Conclusions

Our AI model demonstrated the ability to generate high-quality educational content indistinguishable from that of human experts. Trainees performed similarly on AI-generated and human-generated questions and could not reliably identify the source of the questions. These findings support integrating AI into cardiology education as a tool for augmenting traditional teaching methods and meeting the evolving needs of learners. Embracing AI's potential may contribute to more efficient, personalized, and scalable medical education, ultimately enhancing patient care.

Acknowledgments

We extend our gratitude to the cardiology trainees who participated in this study and to St. Michael's Hospital for access to the Virtual Echo Rounds lectures. We thank Mohammad Ali Khan, MD, for assistance with statistical analysis.

Ethics Statement

The research reported in this article has adhered to all relevant ethical guidelines and principles, including those outlined in the Declaration of Helsinki and applicable institutional policies.

Patient Consent

Not applicable. This study did not involve patient-identifiable data or direct patient involvement. Institutional guidelines did not require individual consent for educational data.

Funding Sources

No funding support was provided by Podcast AI or other entities for this project.

Disclosures

Dr. Chow has a small financial interest in Podcast AI.

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Supplementary Material

To access the supplementary material accompanying this article, visit *CJC Open* at https://www.cjcopen.ca/ and at https://doi.org/10.1016/j.cjco.2025.03.013.